

Name: \_\_\_\_\_

GCSE (1 - 9)

## Expanding and Factorising

Solutions:



These questions are compiled from *released* [Pearson Edexcel GCSE maths exam papers](#)

They contain exam questions from up until the **Summer 2023** exam series

More recent exam papers do exist but are withheld for mock exams

As new exam papers are released these worksheets will gradually be updated

The **most current** version of these worksheets will always be available at: [kennethstafford.co.uk](http://kennethstafford.co.uk)

| Question | Where the question appeared               |
|----------|-------------------------------------------|
| 1        | Edexcel, GCSE, Summer 2017, Paper 2F, Q14 |
| 2        | Edexcel, GCSE, Autumn 2017, Paper 2F, Q24 |
| 3        | Edexcel, GCSE, Autumn 2017, Paper 3F, Q17 |
| 4        | Edexcel, GCSE, Summer 2018, Paper 1F, Q16 |
| 5        | Edexcel, GCSE, Summer 2018, Paper 3F, Q20 |
| 6        | Edexcel, GCSE, Autumn 2018, Paper 2F, Q19 |
| 7        | Edexcel, GCSE, Autumn 2018, Paper 2F, Q26 |
| 8        | Edexcel, GCSE, Summer 2019, Paper 1F, Q16 |
| 9        | Edexcel, GCSE, Autumn 2019, Paper 3F, Q22 |
| 10       | Edexcel, GCSE, Autumn 2020, Paper 1F, Q19 |

## Expanding and Factorising Questions from Past Papers

1

(a) Factorise  $5 - 10m$

.....  
(1)

(b) Factorise fully  $2a^2b + 6ab^2$

.....  
(2)

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**(Total for Question 14 is 3 marks)**

*[Edexcel, GCSE, Summer 2017, Paper 2F, Q14]*

2

(a) Solve  $2x^2 = 72$

.....  
(2)

(b) Expand and simplify  $(2x + 1)(3x - 2)$

.....  
(2)

(c) Factorise  $x^2 + 6x + 9$

.....  
(1)

**(Total for Question 24 is 5 marks)**

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*[Edexcel, GCSE, Autumn 2017, Paper 2F, Q24]*

3

(a) Factorise  $4m + 12$

.....  
(1)

|            |          |         |          |
|------------|----------|---------|----------|
| expression | equation | formula | identity |
| inequality | term     | factor  | multiple |

(b) Choose two words from the box above to make this statement correct.

$5y$  is a ..... in the .....  $3x + 5y$

(2)

**(Total for Question 17 is 3 marks)**

*[Edexcel, GCSE, Autumn 2017, Paper 3F, Q17]*

## Expanding and Factorising Questions from Past Papers

4

$$P = 4x + 3y$$

$$x = 5$$

$$y = -2$$

(a) Work out the value of  $P$ .

.....  
(2)

(b) Expand  $4e(e + 2)$

.....  
(2)

(c) Solve  $3(m - 4) = 21$

$m =$  .....  
(2)

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**(Total for Question 16 is 6 marks)**

*[Edexcel, GCSE, Summer 2018, Paper 1F, Q16]*

5

Expand and simplify  $5(p + 3) - 2(1 - 2p)$

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**(Total for Question 20 is 2 marks)**

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*[Edexcel, GCSE, Summer 2018, Paper 3F, Q20]*

6

(a) Solve  $3(x - 4) = 12$

$x = \dots\dots\dots$   
(2)

(b) Factorise fully  $9b - 3b^2$

$\dots\dots\dots$   
(2)

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**(Total for Question 19 is 4 marks)**

*[Edexcel, GCSE, Autumn 2018, Paper 2F, Q19]*

7

(a) Expand and simplify  $(5x + 2)(2x - 3)$

.....  
(2)

(b) Factorise  $x^2 + 4x + 3$

.....  
(2)

**(Total for Question 26 is 4 marks)**

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*[Edexcel, GCSE, Autumn 2018, Paper 2F, Q26]*



8

(a) Expand  $5(2m - 3)$

.....  
(1)

(b) Factorise  $3n + 12$

.....  
(1)

---

**(Total for Question 16 is 2 marks)**

*[Edexcel, GCSE, Summer 2019, Paper 1F, Q16]*

9

(a) Expand and simplify  $(x + 5)(x - 9)$

.....  
(2)

(b) Factorise fully  $9x^2 + 6x$

.....  
(2)

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**(Total for Question 22 is 4 marks)**

*[Edexcel, GCSE, Autumn 2019, Paper 3F, Q22]*

## Expanding and Factorising Questions from Past Papers

10

(a) Expand  $x(x - 4)$

.....  
(1)

(b) Factorise  $15y - 10$

.....  
(1)

(c) Solve  $7(f - 5) = 28$

$f =$  .....  
(2)

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**(Total for Question 19 is 4 marks)**

*[Edexcel, GCSE, Autumn 2020, Paper 1F, Q19]*